Zirong Chen, Ph.D.

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ACADEMIC APPOINTMENTS & EDUCATION

KTH Royal Institute of Technology, Stockholm, Sweden — Aug 2026 – Present

- **Postdoctoral Fellow** at EECS-CLS (Department of Computing and Learning Systems) and TECoSA (Center for Trustworthy Edge Computing Systems and Applications); working with Dr. David Broman and Dr. Rafia Inam.
- *Research topics*: Trustworthy Assurance, Agentic Memory, Neuro-symbolic Reasoning, Faithfulness Alignment.

Vanderbilt University, Nashville, TN, USA — Sept 2021 – June 2026

- **Doctor of Philosophy (Ph.D.)** in Computer Science; advised by Dr. Meiyi Ma
- *Research Interests*: Trustworthy AI, Natural Language Processing, Reliable LLMs, and Neurosymbolic Reasoning
- *Dissertation*: *Towards Trustworthy NLP-Powered Intelligent Systems for Public Safety*

Georgetown University (**M.S.** in Computer Science), Washington, DC, USA — Sept 2019 – May 2021

Jinling Institute of Technology (**B.E.** in Computer Engineering), Nanjing, PR China — Sept 2015 – May 2019

SELECTED AWARDS & HONORS

IJCAI Student Scholarship Awardee — June 2025

AAAI Student Scholarship Awardee (x2) — Jan 2024, Jan 2025

OpenAI Research Access Program Awardee ($10,000) — Nov 2024

Vanderbilt Student Travel Grant Awardee (x3) — 2023, 2024, 2025

PROPOSAL DEVELOPMENT & CONTRIBUTIONS

Google Society-Centered AI Research Award — Oct 2025

Vanderbilt Catalyst Funds — June 2025

Google Gemma-3 Research Award — May 2025

Google Academic Research Award — Oct 2024

NSF-ReDDDoT (Responsible Design, Development, and Deployment of Technology) — Aug 2024

Vanderbilt GenAI Seeding Grant — Jan 2024

PUBLICATIONS

AAAI-2027 (in submission) **Zirong Chen**, Bingsong Liu, Meiyi Ma. *SimDispatch: Intervention-Sensitive Multi-Agent Simulation System for 9-1-1 Dispatcher Training*.

NeSy-2026 (accepted) **Zirong Chen**, Meiyi Ma. *Reason Popper-ly: Patching In-Context Reasoning with Inductive Logic Programming*.

US Patent (provisional) **Zirong Chen**, Meiyi Ma. *A Signal-Temporal Logic Based Automated Debriefing Approach with Large Language Model Integration*. Filed in July, 2025.

IJCAI-ECAI-2026 (accepted; **Oral Spotlight**) **Zirong Chen**, Hongchao Zhang, Meiyi Ma. *PACE: A Personalized Adaptive Curriculum Engine for 9-1-1 Call-taker Training*.

SmartComp-2026 **Zirong Chen**, Meiyi Ma. *Empowering 9-1-1 Calltaking Training with Generative AI: Experiences and Lessons Learned*, 2026 IEEE International Conference on Smart Computing (SmartComp), Messina, Italy, 2026, pp. 160-167, doi: 10.1109/SmartComp69968.2026.00032.

ICCPS-2026 Ziyan An, Xia Wang, Hendrick Baier, **Zirong Chen**, Abhishek Dubey, Taylor T Johnson, Jonathan Sprinkle, Ayan Mukhopadhyay, Meiyi Ma. *LogiEx: Integrating Formal Logic and LLMs for Explainable Transit Planning*, 2026 ACM/IEEE 17th International Conference on Cyber-Physical Systems (ICCPS), Saint Malo, France, 2026, pp. 88-99, doi: 10.1109/ICCPS68698.2026.00016.

AAAI-2026 David Hanchen Wang, Siwoo Bae, **Zirong Chen**, Meiyi Ma. *Learning with Preserving for Continual*

Multitask Learning. Proceedings of the AAAI Conference on Artificial Intelligence, 40(31), 26178–26186. https://doi.org/10.1609/aaai.v40i31.39821.

AAMAS-2025 Ziyan An, Xia Wang, Hendrick Baier, **Zirong Chen**, Abhishek Dubey, Taylor T Johnson, Jonathan Sprinkle, Ayan Mukhopadhyay, Meiyi Ma. *Combining LLMs with a Logic-Based Framework to Explain MCTS*. Published (39.9%) at AAMAS-2025, Detroit, MI, pp. 2405–2407.

IJCAI-2025 **Zirong Chen**, Ziyan An, Jennifer Reynolds, Kristin Mullen, Stephen Martini, Meiyi Ma. *LogiDebrief: A Signal-Temporal Logic Based Automated Debriefing Approach with Large Language Models Integration*. Published and **Selected Oral Spotlight Presentation** at IJCAI-2025 (**13%**), Montreal, Canada, 9582–9590. https://doi.org/10.24963/ijcai.2025/1065.

AAAI-2025 **Zirong Chen**, Elizabeth Chason, Noah Mladenovski, Erin Wilson, Kristin Mullen, Stephen Martini, Meiyi Ma. *Sim911: Towards Effective and Equitable 9-1-1 Dispatcher Training with an LLM-Enabled Simulation*. Published (**18.9%**) and **Selected Oral Spotlight Presentation (4.9%)** at AAAI-2025, Philadelphia, PA, 39(27), 27896-27904. https://doi.org/10.1609/aaai.v39i27.35006

AAAI-2024 **Zirong Chen**, Xutong Sun, Yuanhe Li, Meiyi Ma. *Auto311: A Confidence-Guided Automated System for Non-emergency Calls*. Published (**20.1**%) and **Selected Oral Spotlight Presentation (6%)** at AAAI-2024, Vancouver, Canada, 38(20), 21967-21975. https://doi.org/10.1609/aaai.v38i20.30199.

Pervasive and Mobile Computing 2023 **Zirong Chen**, Isaac Li, Haoxiang Zhang, Sarah Preum, John A Stankovic, Meiyi Ma. *CitySpec with shield: A secure intelligent assistant for requirement formalization*. Invited journal extension at Elsevier's Pervasive and Mobile Computing (2023),Volume 92, 2023, 101802, ISSN 1574-1192, https://doi.org/10.1016/j.pmcj.2023.101802.

SmartComp-2022 **Zirong Chen**, Isaac Li, Haoxiang Zhang, Sarah Preum, John A Stankovic, Meiyi Ma. *CitySpec: An intelligent assistant system for requirement specification in smart cities*. Published and **Best Paper Finalist** at IEEE SmartComp 2022, Helsinki, Finland, pp. 32-39, doi: 10.1109/SMARTCOMP55677.2022.00020.

SmartComp-2022 **Zirong Chen**, Isaac Li, Haoxiang Zhang, Sarah Preum, John A Stankovic, Meiyi Ma. *An intelligent assistant for converting city requirements to formal specification*. Published as Demo Paper at IEEE SmartComp 2022, Helsinki, Finland, pp. 174-176, doi: 10.1109/SMARTCOMP55677.2022.00043.

RESEARCH PROJECTS

Angie: AI-Driven Training System for 911 Call-takers in Emergency Responses Dec 2023 - Present
Integrating AI with real-world applications for a more efficient and effective emergency response training solution

- Led AI research project with Metro Nashville Emergency Communications, IT Services, and Government. Secured research grants from *OpenAI*, *Google*, *NSF(ReDDDoT)*, and *Vanderbilt*.
- Served as **major contributor** and AI tech vendor, developing pioneering emergency response solution with sub-systems: *PACE*, *LogiDebrief*, *Sim911*, and *Auto311*. Deployed across 400+ workstations with 200+ active users, serving Metro Nashville for 14+ months.

SimDispatch: Topology-Bounded Multi-Agent Dispatching Simulation Jan 2026 - Present
A multi-agent role playing with stateful representation of system events with real-world clocks

- Proposed two layered transitions of multi-agent-role playing for 9-1-1 dispatching training: control transitions and acting transitions. Used topology to bound underlying event states and seggrated LLM agents behaviors with persona and ReAct-like prompts.
- Deployed at MNDEC and under active refinement since Nov 2025. Saving 11 minutes per simulation (under estimation). Making dispatcher training easier and more accessible.

PACE: A Personalized and Apative Curriculum Engine for 9-1-1 Training Nov 2025 - Present
Formulate 9-1-1 call-taking knowledge with structural representation, accomodate to different learning patterns

- Formulate 9-1-1 call-taking curriculum adaptation as a recommendation problem with both explicit and implicit objectives, modeling with contextual bandit.
- Estimate and capture trainee's learning patterns from cognitive and educational perspectives over training history, accomodating to different training needs and optmizing the training gains.

LogiDebrief: An STL-based Automated 911 Call Debriefing System with LLMs Nov 2024 - Present
Disentangle overly complicated prompts into logic-enforced specifications with modularized use of LLMs

- Developed signal temporal logic (STL) framework integrated with LLMs to rigorously analyze and debrief historical 911/311 calls, maintaining accuracy across extended context windows. Through extensive real-world

testing, LogiDebrief achieves over 95% accuracy without performance degradation on long contexts.

- Successfully deployed at local emergency response center for both daily operations and call-taker training programs. The system has automated analysis of over 1,701 emergency calls, resulting in 311+ hours of operational time savings.

Sim911: An LLMs-based Simulation System for Call-taker Training Mar 2024 - Present
An effective and reliable workflow for LLMs role-play simulation under emergency responses

- Developed context-aware generation and validation framework using CoT, RAG, and few-shot prompting with LLM agents to ensure realistic and inclusive simulations (addressing underserved groups). Framework incorporates automatic correction based on human feedback for continuous improvement.
- In real-world training programs, 90%+ participants rated Sim911 as comparable or superior to human-led simulations. Framework outperforms baseline by 62% in authenticity benchmarks.
- Deployed in local emergency response center's call-taker training programs. Generated 600+ simulated calls with 40+ hours of active engagement, saving 80+ hours of instructor time.

Auto311: A Confidence-Guided Automated System for Non-emergency Calls Jan 2023 - Present
Integrating AI with real-world applications for an automated call handling system

- Collaborated with Metro Nashville Department of Emergency Communications, Metro Nashville Information Technology Services, and Metro Nashville Government.
- Designed and developed an automatic call handling system for non-emergency calls based on confidence measurement with deep neural networks. Outperformed naïve LLMs calls with 94.49% acc under emulation. Implemented baseline models like Transformers.

INDUSTRY EXPERIENCE

AI Research Fellow Intern, Epiq AI Labs (formerly LAER AI) – New York, NY May – Aug 2025

- Enhanced AI eDiscovery Assistant^{TM} with **DeepResearch** capabilities through agentic workflow design, optimized for eDiscovery, document review, and complex legal scenarios.
- Created **BREEW** (**B**ayesian **R**easoning via **E**ntity-**E**vidence **W**eaving) benchmark that leverages concept decomposition and entity-level evidence surfacing to ensure logical coherence and rigorous concept validation.
- Built **LINT** (**L**earning via **IN**teractive **T**utoring) testing environment featuring simulated user agents with restorable states. The system evaluates agent performance through controlled conversational interactions using probabilistic finite state machines and configurable personas.

Data Scientist Intern, **National Basketball Association** (NBA) – New York, NY May - Aug, 2021

- Analyzed user feedback using neural networks and NLP tools to develop data-driven fan engagement strategies.
- Enhanced marketing campaign effectiveness by applying statistical ML models to optimize user experience and advertisement targeting.
- Developed real-time data analysis pipelines with sequential neural networks to predict streaming bandwidth requirements, optimizing infrastructure costs and user experience.

ML Engineer Intern, **Chinese Academy of Sciences** – Nanjing, China Dec 2018 - May 2019

- Developed projection algorithms based on vectorized path trajectories and distance metrics to improve GPS accuracy in dense urban areas.
- Created dynamic routing algorithms to optimize urban planning and navigation tasks.
- Contributed to a high-impact, data-driven project involving predictive modeling and real-time decision-making for urban management applications.

ACADEMIC SERVICES

Program Committee at AAAI: Main Technical, AAAI: Social Impact. 2024 - Present

Conference Reviewer at NeurIPS, ICLR, IJCAI, ICCPS, SenSys, BuildSys. 2023 - Present

Journal Reviewer at ACM Transactions on Computing for Healthcare, Pervasive and Mobile Computing. 2023 - Present

Course Co-designer at Vanderbilt University: CS-5891 (Explainable AI, LLMs interpretability). 2026

Teaching Assistant at Vanderbilt University: CS-4289 (Web-based System Development), CS-6315 (Automated Verification), CS-8395 (Deep Learning), CS-5260 (Artificial Intelligence), CS-8395 (AI in Cyber Physical Systems). 2021 - 2025

Teaching Assistant at Georgetown University: COSC-576 (Neural Networks), ANLY-590 (Deep Learning), GBUS-401 (Big Data in Business, Economy, Society). 2019 - 2021

INVITED TALKS & LECTURES

Invited Talk: "Building Trustworthy AI for Public Safety," TECoSA, KTH. 2026
Invited Talk: "Towards Trustworthy Intelligence," EECS-CLS, KTH. 2026
Invited Talk: "Empowering 911 Training with Trustworthy AI," by Dr. Rafia Inam at Ericsson AB, Sweden. 2026
Invited Talk: "Symbolic AI for Public Safety," by Dr. Luc De Raedt at KU Leuven. 2025
Invited Talk: "GenAI in 911 Training," by Dr. Eshan Hoque at Uni. of Rochester. 2025
Invited EduAI Demo at Vanderbilt LIVE Learning Innovation Incubator (K-12). 2024 - 2025
Guest Lecture at Vanderbilt (Automated Verification): "Intro to Formal Logic." 2024
Guest Lecture at Vanderbilt (AI in CPS): "Large Language Models." 2022 - 2025
Guest Lecture at Vanderbilt (Artificial Intelligence): "Natural Language Processing." 2022 - 2024
Guest Lecture at NBA (Data Science): "Autoencoders in Applications." 2020
Guest Lecture at NBA (Intelligent Marketing): "Deep Learning From Ground Up." 2020